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Communications Infrastructure

Data Center Electrification: Power Architecture Must Evolve to Support AI; Grid Integrity at Risk from Large Loads Without BTM Solutions; Expect More Fiber Builds

Communications Infrastructure

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We recently hosted an industry expert call with Al Cioffi, CTO of 4GM Management Group and a leading expert in data center power systems and architecture, for a discussion on how the rapidly scaling AI infrastructure is reshaping the approach to data center electrification. The conversation covered the implications of gigawatt-scale facilities and increasingly dynamic loads on grid stability, the shift toward behind-the-meter (BTM) generation, and the emerging role of pre-fabrication and modular designs. The overarching takeaway was that the largest AI campuses increasingly need to be designed as power systems first and compute facilities second, creating greater demand for integrated engineering and equipment solutions. We continue to see Digital Realty Trust (DLR), Equinix (EQIX), and Csquare (CSQR) as traditional winners in the capacity build out for cloud and AI workloads, alongside HPC providers Riot Platforms (RIOT) and Cipher Digital (CIFR), and neocloud provider IREN (IREN). Finally, we see both Dycom (DY) and Uniti (UNIT) as beneficiaries of large-scale hyperscale related fiber builds. A link to the full call was posted on JPMM and can be found here.

- **Gigawatt-scale AI campuses fundamentally change historical assumptions about grid loads.** In the past, traditional loads on the grid were small in comparison to the generation facilities. Data centers of 20-40 MW were considered large, but even that was overshadowed by generation in the hundreds of MW. Today, GPU rack densities and AI scaling laws have resulted in demand for GW scale loads that rival the power capacity of the generation itself. To put this into perspective, a standard AP1000 nuclear reactor generates ~1 GW of capacity, while many hyperscalers have announced plans for multi-GW campuses (e.g., Stargate, Prometheus, Rainer). Additionally, the synchronous use of GPUs during training runs can cause significant changes in power consumption, with data centers going from high utilization close to peak capacity to <50% within milliseconds. These changes can create grid disturbances that are difficult to manage, particularly as conventional generation with physical inertia (coal, gas, nuclear, and hydro) is retired and replaced by inverter-based renewable resources (solar, wind). Grid disturbances could lead to outages, damage, and other grid disruptions, which would add to existing negative data center sentiment widely expressed today. From Cioffi's perspective, the largest training campuses will increasingly need to bring their own generation and stability rather than rely exclusively on a utility connection. This favors BTM solutions using dispatchable generation, most likely natural gas or nuclear for large sites, while inference capacity should be easier to accommodate because of the distributed nature of the compute and smaller scale of capacity needed.
- **Cloud era is different than the AI era given power densities; fiber is difficult to build but a faster solution than power.** In the past, cloud servers were 3-10 kW/server, but with AI servers reaching 40-80-100+ kW/server, the capacity needs for each data center building or campus have increased dramatically. As discussed above, AI training runs can drive large swings in

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power usage. While these swings present grid challenges, we see capacity and utilization associated with AI inference as much more predictable – similar to cloud workloads, but the fundamental power density difference means even inference swings could be too difficult for the grid to handle. Going forward, grid architecture needs to be augmented with innovative BTM, electrical distribution, and power storage solutions. Natural gas and, secondarily, nuclear appear to be the most likely "island" or "off-grid" solutions for HPC data centers given the availability of equipment and fuel (nat gas) and reliability and scale of capacity (nuclear). We think it is easier to develop independent power data centers near natural gas locations on or near pipelines and bring fiber to those locations rather than solving for fiber first.

- **Behind-the-meter generation changes the entire electrical architecture, not just the source of power.** Campuses opting for a BTM solution are choosing an approach that must provide grid reliability and redundancy without a grid connection. While many of these projects see BTM as a bridge solution to a formal utility connection, in the interim the on-site generation must take on this responsibility. Developers may avoid a conventional utility substation, but they still require electrical protection, transformation, switchgear, and distribution infrastructure designed as an integrated system. The result is higher build costs for data centers opting for BTM solutions and a higher degree of engineering expertise to pull off these new solutions.
- **Modularization should help compress construction timelines and address labor shortages.** While delays related to NIMBYism or lengthy permitting queues are bottlenecks to bringing new capacity online, the construction process itself is being reimagined by the industry. Standardization of the design and construction process can help speed up the time to delivery and modularization has emerged as a creative solution to accelerating RFS dates. Suppliers with deep engineering capabilities and experience with power are starting to provide end-to-end solutions that integrate multiple pieces of equipment into a single product. Beyond a speed advantage, modularization is helping address current labor issues. By eliminating some complexities related to the deployment of electrical systems, electricians and engineers are freed up to work on other portions of the data center build.

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